

Experiment NO: 1

Experiment Name: Familiarization with MDA-8086 Microprocessor Kit and Its Operation in Machine Codes.

Objectives:

1. To become familiar with the MDA-8086 microprocessor kit and its components.
2. To learn how to enter and execute machine codes using the MDA-8086 kit.
3. To understand the basic use of 8086 instructions such as `MOV` and `ADD`.

Equipment:

1. MDA-8086 Microprocessor Kit
2. Power Supply
3. Emu8086 Software

Description:

The MDA-8086 is a training kit based on the Intel 8086 microprocessor. It provides a keyboard for entering machine codes and an LCD display for showing the results. In this experiment, machine codes are entered into the kit and executed to perform register operations and addition.

Procedure:

1. Turn on the **MDA-8086** kit using the required power supply.
2. Use the kit keyboard to enter the required machine codes.
3. Load **1234H** into the **AX** register.
4. Load **4321H** into the **BX** register.
5. Execute the **ADD AX, BX** instruction.
6. Check the register values using the **Reg** option.
7. Compare the obtained result with the result from Emu8086 software.



Fig: Microprocessor Trainer-SB-8086

Example: Two Address Addition

The assembly instructions used were:

```
MOV AX, 1234H
MOV BX, 4321H
ADD AX, BX
```

The machine-code steps performed in the kit were:

Step	Input/Operation	Address/Value
1	AD	—
2	:	—
3	DA	—
4	B8 +	1001
5	34 +	1002
6	12 +	1003
7	STP	—
8	Reg	AX = 1234H
9	AD	—
10	:	—
11	DA	—
12	BB +	1004
13	21 +	1005
14	43 +	1006
15	STP	—
16	Reg	BX = 4321H
17	AD	—
18	:	—
19	DA	—
20	03 +	—
21	C3 +	—
22	F4 + C	—
23	STP	—
24	Reg	AX = 1234H, BX = 4321H

Input/Output:

Input:

```
MOV AX, 1234H
MOV BX, 4321H
ADD AX, BX
```

Machine Codes:

```
MOV AX, 1234H → B8 34 12
MOV BX, 4321H → BB 21 43
ADD AX, BX      → 03 C3
```

Output:

```
AX = 5555H
BX = 4321H
```

Calculation: Final Answer:

```
  1234H
+ 4321H
-----
  5555H
```

Discussions:

In this experiment, we learned how to operate the MDA-8086 kit in machine-code mode and perform a basic addition operation. The machine codes were also checked using Emu8086 software. The result obtained from the kit matched the expected result, so the experiment was successfully completed.